

DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES E INFORMÁTICA

MESTRADO EM ENGENHARIA DE COMPUTADORES E TELEMÁTICA

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MODELAÇÃO E DESEMPENHO DE REDES E SERVIÇOS

PRACTICAL GUIDE NO. 2

Task 1

Consider the discrete event simulator, implemented in the provided MATLAB function *Sim2Flows*, to estimate the performance of a point-to-point link supporting 2 packet flows. The simulator considers a link of capacity C (in Mbps) and a queue of size f (in Bytes) with a FIFO (First-In-First-Out) scheduling discipline.

The packet Flow 1 is characterized by: (i) a packet arrival following a Poisson process with average λ_1 pps and (ii) a random packet size between 64 and 1518 bytes with the probabilities: 19% for 64 bytes, 23% for 110 bytes, 17% for 1518 bytes and an equal probability for all other values (i.e., from 65 to 109 and from 111 to 1517).

The packet Flow 2 is characterized by: (i) a packet arrival following a Poisson process with average λ_2 pps and (ii) a random packet size between 110 and 130 bytes (all values with equal probability).

Input parameters of *Sim2Flows*:

- λ – vector with the packet rate of each flow, in pps
- C – link capacity, in Mbps
- f – queue size, in Bytes
- P – total number of transmitted packets of a simulation run

Performance parameters estimated by *Sim2Flows*:

- PL – vector with the Packet Loss (in %) of each flow
- APD – vector with the Average Packet Delay (in milliseconds) of each flow
- MPD – vector with the Maximum Packet Delay (in milliseconds) of each flow
- TT – vector of the Transmitted Throughput (Mbps) of each flow

Stopping criterion of *Sim2Flows*:

- Time instant when the link finishes the transmission of the Pth packet (P is the 4th input parameter); in *Sim2Flows*, the queued packets at the end of the simulation do not count for performance estimation.

1.a. Develop a MATLAB script to run *Sim2Flows* 50 times with a stopping criterion of P = 10.000 at each run and to compute the estimated values and the 90% confidence intervals of all performance parameters when λ_1 = 1500 pps, λ_2 = 1000 pps, C = 10 Mbps and f = 1.000.000 Bytes (~1 MByte). Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.68e+00 +- 8.39e-02
  Max. Packet Delay (ms) = 1.54e+01 +- 8.87e-01
  Throughput (Mbps)      = 7.42e+00 +- 3.35e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.26e+00 +- 7.94e-02
  Max. Packet Delay (ms) = 1.49e+01 +- 8.90e-01
  Throughput (Mbps)      = 9.61e-01 +- 3.96e-03
```

1.b. Repeat experiment **1.a** now considering f = 10.000 Bytes (~10 KBytes) Justify the differences between these results and the results of experiment **1.a**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 2.47e+00 +- 7.41e-02
  Avg. Packet Delay (ms) = 1.64e+00 +- 1.25e-02
  Max. Packet Delay (ms) = 5.01e+00 +- 2.24e-02
  Throughput (Mbps)      = 7.07e+00 +- 2.23e-02
Flow 2:
  Packet Loss (%)        = 3.58e-01 +- 3.32e-02
  Avg. Packet Delay (ms) = 1.31e+00 +- 1.29e-02
  Max. Packet Delay (ms) = 4.78e+00 +- 2.98e-02
  Throughput (Mbps)      = 9.57e-01 +- 3.21e-03
```

1.c. Repeat experiment **1.a** now considering f = 2.000 Bytes (~2 KBytes) Justify the differences between these results and the results of experiments **1.a** and **1.b**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 9.71e+00 +- 1.03e-01
  Avg. Packet Delay (ms) = 9.20e-01 +- 3.05e-03
  Max. Packet Delay (ms) = 2.70e+00 +- 1.09e-02
  Throughput (Mbps)      = 6.10e+00 +- 1.81e-02
Flow 2:
  Packet Loss (%)        = 1.37e+00 +- 5.78e-02
  Avg. Packet Delay (ms) = 6.44e-01 +- 3.14e-03
  Max. Packet Delay (ms) = 2.58e+00 +- 1.54e-02
  Throughput (Mbps)      = 9.44e-01 +- 3.58e-03
```